

A

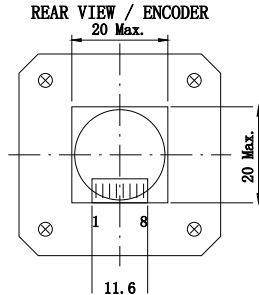
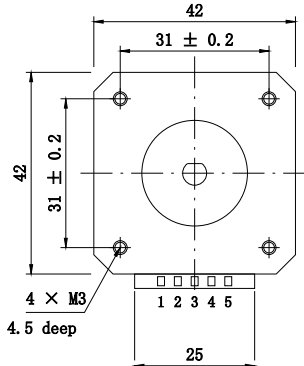
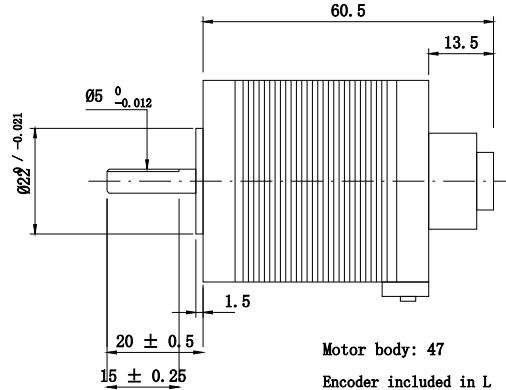
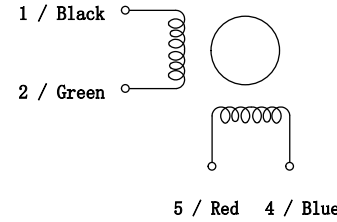
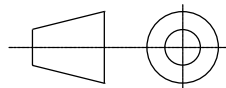
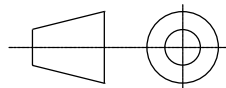
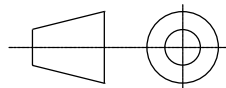
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| <div><div><p>MODEL REFERENCE</p><div>PKP245D23A2-R2FL</div><div>PKP SERIES / ENCODER</div></div><p>Sources: Oriental Motor product catalog 2018/2019, pp. A-65, A-79; HM-7475-4E, pp. 18, 24, 26.<br/>Encoder: HM-7439-3JE, English page 2.<br/>Dimensions / mass: PKP Series catalog 2020/2021, p. 29, drawing B1323 (manufacturer supplement).</p><table><tr><th colspan="4">ENCODER CONNECTION / R2FL</th></tr><tr><th>PIN</th><th>LEAD COLOR</th><th>SIGNAL</th><th>FUNCTION</th></tr><tr><td>1</td><td>Black</td><td>GND</td><td>Supply ground</td></tr><tr><td>2</td><td>Red</td><td>A+</td><td>Phase A positive</td></tr><tr><td>3</td><td>Brown</td><td>A-</td><td>Phase A negative</td></tr><tr><td>4</td><td>Green</td><td>B+</td><td>Phase B positive</td></tr><tr><td>5</td><td>Blue</td><td>B-</td><td>Phase B negative</td></tr><tr><td>6</td><td>Yellow</td><td>Z+</td><td>Index positive</td></tr><tr><td>7</td><td>Orange</td><td>Z-</td><td>Index negative</td></tr><tr><td>8</td><td>White</td><td>Vcc</td><td>Supply +5 V</td></tr></table><p>Mating connector (encoder):<br/>Molex 51021-0800</p><p>Contacts: 50079-8X00</p><p>Cable: AWG28 - 26</p><p>Lead colors refer to the Oriental Motor cable.</p></div> <div><table><tr><th colspan="2">ENCODER R2FL</th></tr><tr><td>RESOLUTION</td><td>400 P/R</td></tr><tr><td>OUTPUT</td><td>Line driver</td></tr><tr><td>SUPPLY</td><td>5 V ±10%</td></tr><tr><td>CURRENT</td><td>≤30 mA</td></tr><tr><td>RESPONSE</td><td>200 kHz</td></tr></table></div> |                        |     |             |   |   | ENCODER CONNECTION / R2FL   |                                               |                                                                                       |                  | PIN                                                      | LEAD COLOR | SIGNAL                                      | FUNCTION   | 1                   | Black      | GND           | Supply ground | 2                       | Red | A+ | Phase A positive | 3                       | Brown | A-                         | Phase A negative | 4                       | Green | B+ | Phase B positive | 5                                                                                     | Blue | B-          | Phase B negative | 6 | Yellow | Z+ | Index positive | 7 | Orange | Z-         | Index negative | 8 | White | Vcc | Supply +5 V | ENCODER R2FL |  | RESOLUTION  | 400 P/R | OUTPUT | Line driver | SUPPLY | 5 V ±10%                                      | CURRENT | ≤30 mA     | RESPONSE                                                 | 200 kHz | <table><tr><th colspan="2">CHARACTERISTICS</th></tr><tr><th>ITEMS</th><th>SPECIFICATIONS</th></tr><tr><td>Rated Voltage</td><td>DC 2.3 V</td></tr><tr><td>Rated Current</td><td>2.3 A / phase</td></tr><tr><td>Step Angle</td><td>1.8°</td></tr><tr><td>Phase</td><td>2 (bipolar)</td></tr><tr><td>Resistance per phase</td><td>1.0 Ω / phase</td></tr><tr><td>Inductance per phase</td><td>2.2 mH / phase</td></tr><tr><td>Holding Torque</td><td>660 mN • m (max.)</td></tr><tr><td>Insulation Resistance</td><td>≥100 MΩ at 500 VDC</td></tr><tr><td>Insulation Class</td><td>130 (B)</td></tr><tr><td>Rotor Inertia</td><td>73 g • cm<sup>2</sup></td></tr><tr><td>Weight</td><td>0.38 kg</td></tr></table> <p>Stop position accuracy: ±0.05°<br/>Full step, no load. Mass includes the encoder.</p> <p>Wiring Diagram:<br/>Motor: bipolar, Model A connector</p>  <p>Pin 3: not connected.<br/>Coil A: Black - Green. Coil B: Red - Blue.<br/>Mating housing: Hirose MDF97-5S-3.5C<br/>HM-7475-4E also lists MDF97A-5S-3.5C.</p> |  | CHARACTERISTICS  |            | ITEMS | SPECIFICATIONS  | Rated Voltage | DC 2.3 V | Rated Current | 2.3 A / phase | Step Angle | 1.8° | Phase | 2 (bipolar) | Resistance per phase | 1.0 Ω / phase | Inductance per phase | 2.2 mH / phase | Holding Torque | 660 mN • m (max.) | Insulation Resistance | ≥100 MΩ at 500 VDC | Insulation Class | 130 (B) | Rotor Inertia | 73 g • cm <sup>2</sup> | Weight | 0.38 kg |
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| ENCODER R2FL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                        |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| RESOLUTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 400 P/R                |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| OUTPUT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Line driver            |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| SUPPLY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 5 V ±10%               |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| CURRENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ≤30 mA                 |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| RESPONSE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 200 kHz                |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| CHARACTERISTICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                        |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| ITEMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SPECIFICATIONS         |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
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| Holding Torque                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 660 mN • m (max.)      |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| Insulation Resistance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ≥100 MΩ at 500 VDC     |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
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| Weight                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.38 kg                |     |             |   |   |                             |                                               |                                                                                       |                  |                                                          |            |                                             |            |                     |            |               |               |                         |     |    |                  |                         |       |                            |                  |                         |       |    |                  |                                                                                       |      |             |                  |   |        |    |                |   |        |            |                |   |       |     |             |              |  |             |         |        |             |        |                                               |         |            |                                                          |         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                  |            |       |                 |               |          |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
| <table><tr><th colspan="2">Dimensions refer to drawing</th><th colspan="2">Tolerance sheet</th><th colspan="2" rowspan="3">ORIENTAL MOTOR<br/>PKP SERIES / WITH ENCODER</th></tr><tr><td>Unit: mm</td><td>Scale: nts</td><td colspan="2">General tolerances:</td></tr><tr><td colspan="2">Product Code:</td><td colspan="2">Not specified in source</td></tr><tr><td colspan="2"></td><td colspan="2">Model: PKP245D23A2-R2FL</td><td colspan="2">Use stated dimensions only</td><td colspan="2">Model: PKP245D23A2-R2FL</td></tr><tr><td colspan="2"></td><td colspan="2" rowspan="3"></td><td colspan="2">Designed by</td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="2">Checked by</td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="2">Approved by</td><td colspan="2"></td></tr><tr><td>A</td><td colspan="2">Prepared from Oriental Motor source documents</td><td>2026.09.06</td><td colspan="2">Mechanical source: B1323<br/>PKP catalog 2020/2021, p. 29</td><td colspan="2" rowspan="2">Drawing No.: PKP245D23A2-R2FL</td><td>Page 1 of 1 Page</td><td>Version: A</td></tr><tr><td>REV.</td><td colspan="2">Revision Record</td><td>Date</td><td colspan="6"></td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                        |     |             |   |   | Dimensions refer to drawing |                                               | Tolerance sheet                                                                       |                  | ORIENTAL MOTOR<br>PKP SERIES / WITH ENCODER              |            | Unit: mm                                    | Scale: nts | General tolerances: |            | Product Code: |               | Not specified in source |     |    |                  | Model: PKP245D23A2-R2FL |       | Use stated dimensions only |                  | Model: PKP245D23A2-R2FL |       |    |                  |  |      | Designed by |                  |   |        |    |                |   |        | Checked by |                |   |       |     |             |              |  | Approved by |         |        |             | A      | Prepared from Oriental Motor source documents |         | 2026.09.06 | Mechanical source: B1323<br>PKP catalog 2020/2021, p. 29 |         | Drawing No.: PKP245D23A2-R2FL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | Page 1 of 1 Page | Version: A | REV.  | Revision Record |               | Date     |               |               |            |      |       |             |                      |               |                      |                |                |                   |                       |                    |                  |         |               |                        |        |         |
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